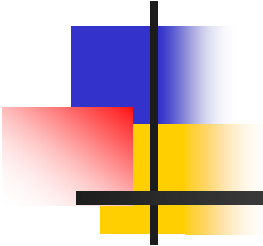


Advanced Manufacturing
Technology & Prototyping
MDP494_UG:2018



Lecture 3

Introduction to CAD/CAM/CIM/CAE

DR. Moustafa Mahmoud

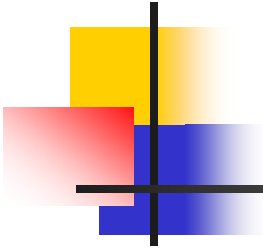
Ph. D. Eng.

Design & Production Dept.,

Faculty of Engineering, Ain Shams University

m_m_sayed@eng.asu.edu.eg

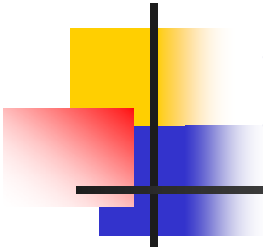
Dr- Mostafa Mahmoud



Introduction

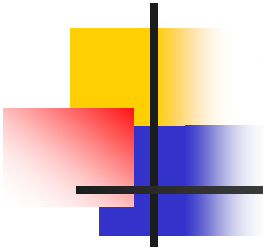
In today's global competition, industries cannot survive unless they introduce new products or existing ones with:

- ***Better quality***
- ***Lower cost***
- ***Shorter lead time***



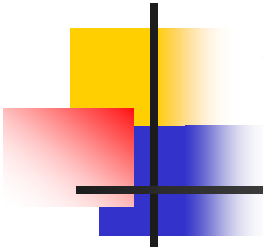
Definition 1

- **Computer-aided design (CAD)** is the use of computer systems to assist in the **creation, modification, analysis, or optimization** of a design.



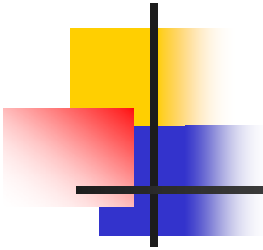
Definition 2

- **Computer-aided manufacturing (CAM)** is the use of computer systems to **plan, manage, and control the operations** of a manufacturing plant through direct or indirect computer interface with plant's resources.



Definition 2, cont.

- 3D CAD data can be read by CAM software which takes 3D data & CNC machine parameters as inputs and delivers a tool path that cuts metal as per the part designs.
- The tool path so generated can be simulated on the screen to evaluate tool gauging so that the machining run is perfect.

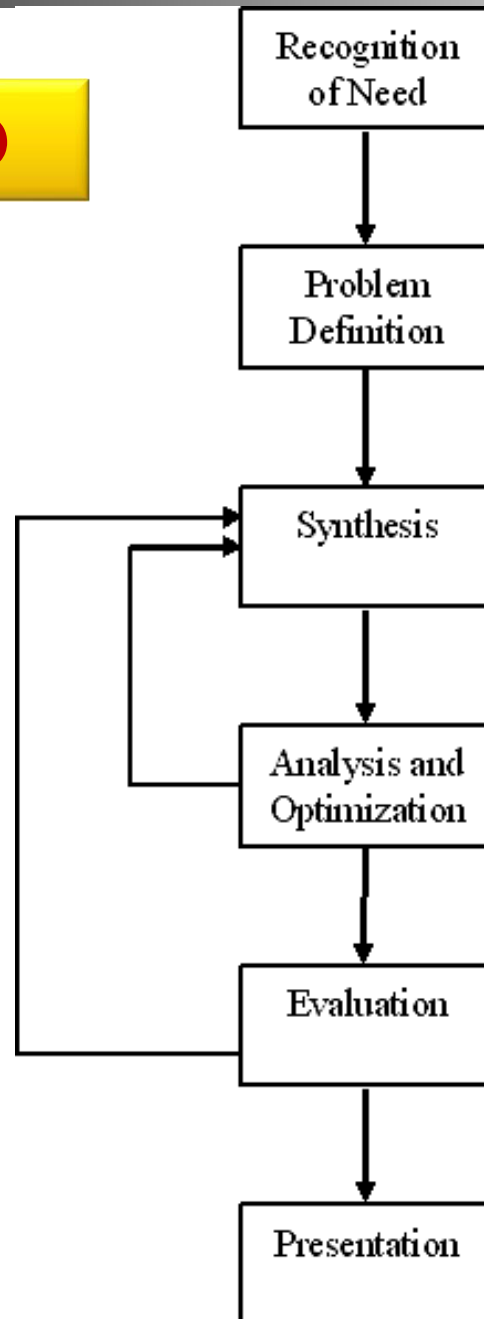


Need for CAD/CAM

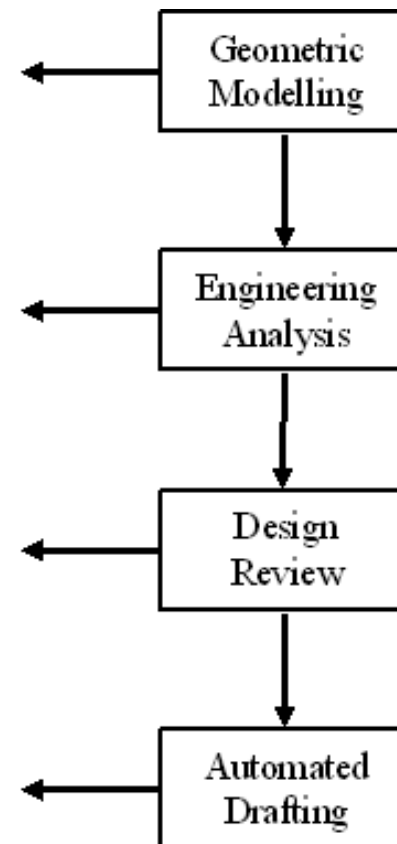
- To increase productivity of the designer
- To improve quality of the design
- To improve communications
- To create a manufacturing database
- To create and test tool paths and optimize them
- To help in production scheduling and MRP models
- To have effective shop floor control

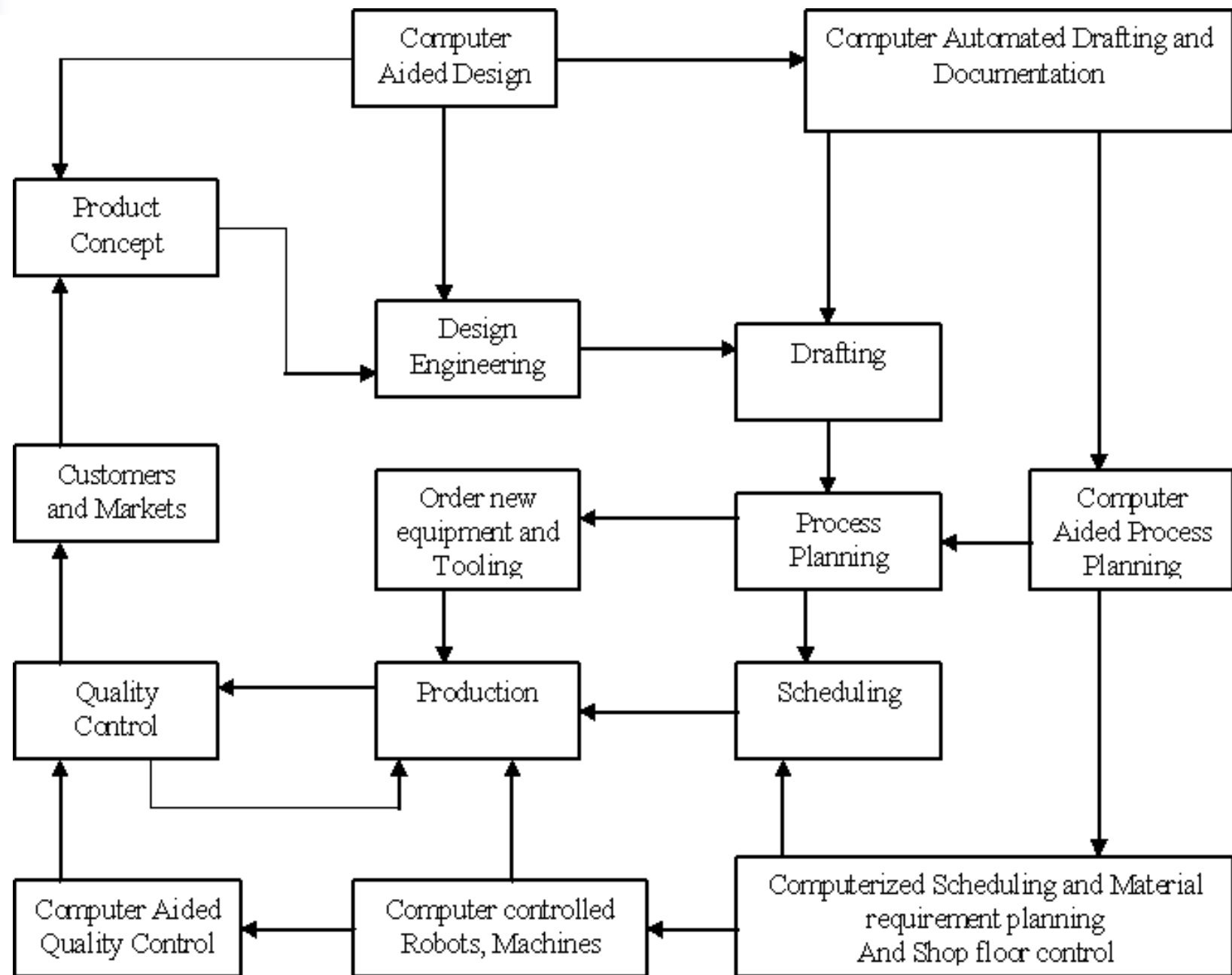
The Design Process : Then and Now

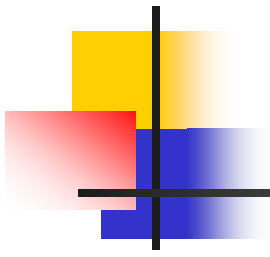
Before CAD



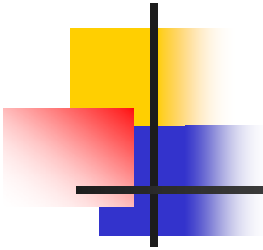
After CAD







EXERCISE_PART A (15-20 MINS)



Exercise 1

- **Discuss how CAD/CAM helps in Lean Manufacturing? Elaborate on any one aspect.**

What is lean manufacturing?

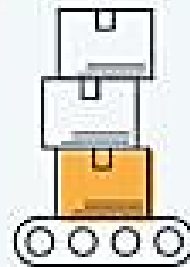
- Lean manufacturing is a production philosophy that aims to **minimize waste** while maximizing productivity and customer satisfaction.
- “Going lean” may involve implementing a number of different productivity-driving methods and tools to achieve continuous improvement.

Hint, cont.

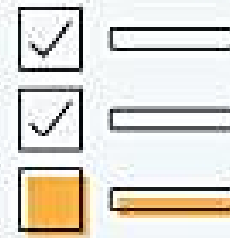
8 wastes of lean manufacturing



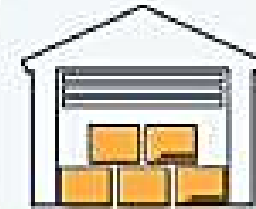
Unnecessary
transportation



Overproduction



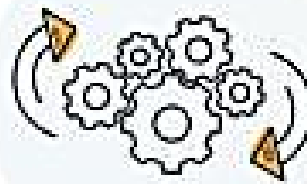
Unnecessary
motion



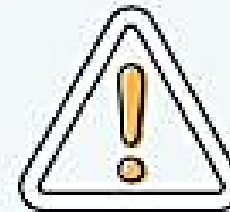
Excess
inventory



Waiting
(idle time)



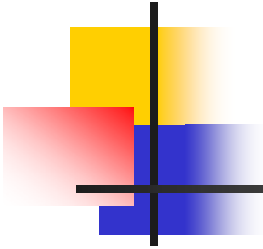
Overprocessing



Defects

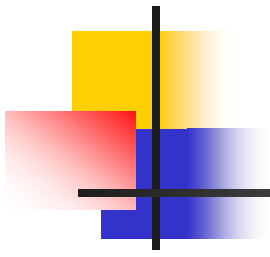


Unutilized
talent

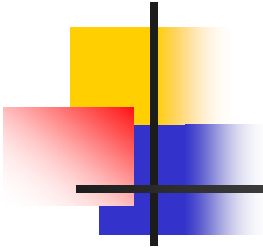


Exercise 2

- **What advantages does CAD/CAM approach offer in NC Programming?**

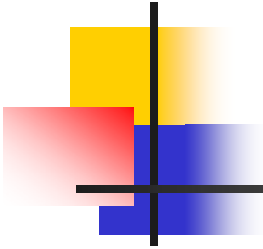


PRODUCT CYCLE (DESIGN AND MANUFACTURING)

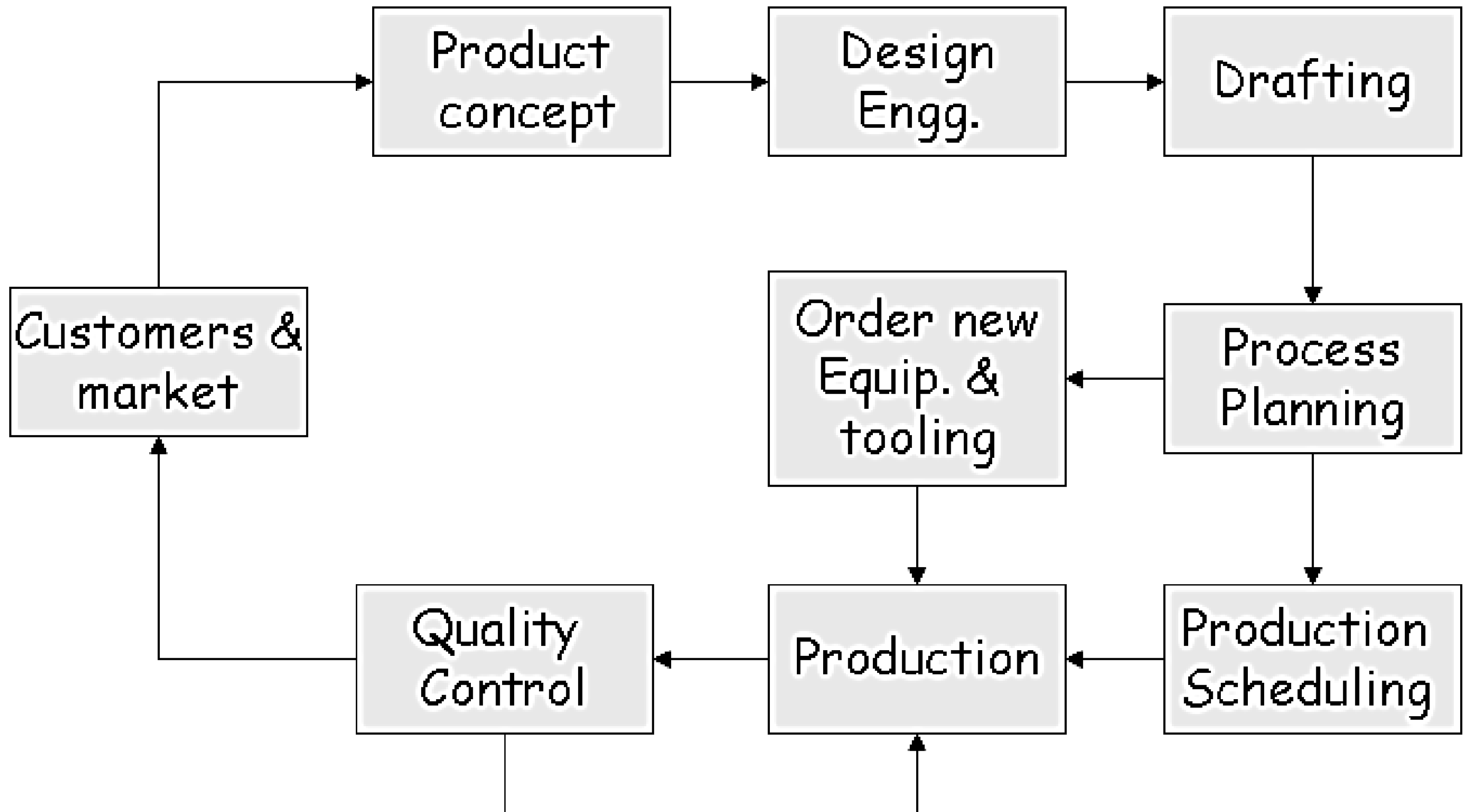


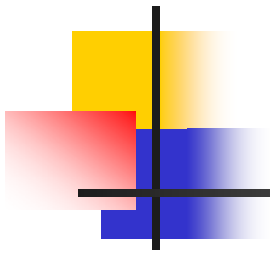
Product Cycle (Design and Manufacturing)

- Driven by *customers* and *markets* which demand the product
- In some cases, design functions are performed by the customer and the product is manufactured by a **different firm**.
- In other cases, **design** and **manufacturing** is accomplished by the **same firm**

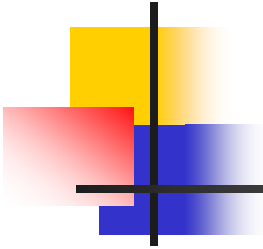


Product Cycle (Design and Manufacturing)



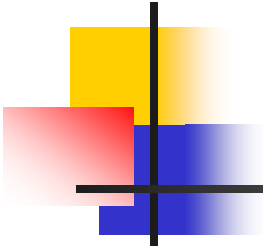


PRODUCT CYCLE REVISED WITH **CAD/CAM** OVERLAID



Product Cycle revised with CAD/CAM overlaid

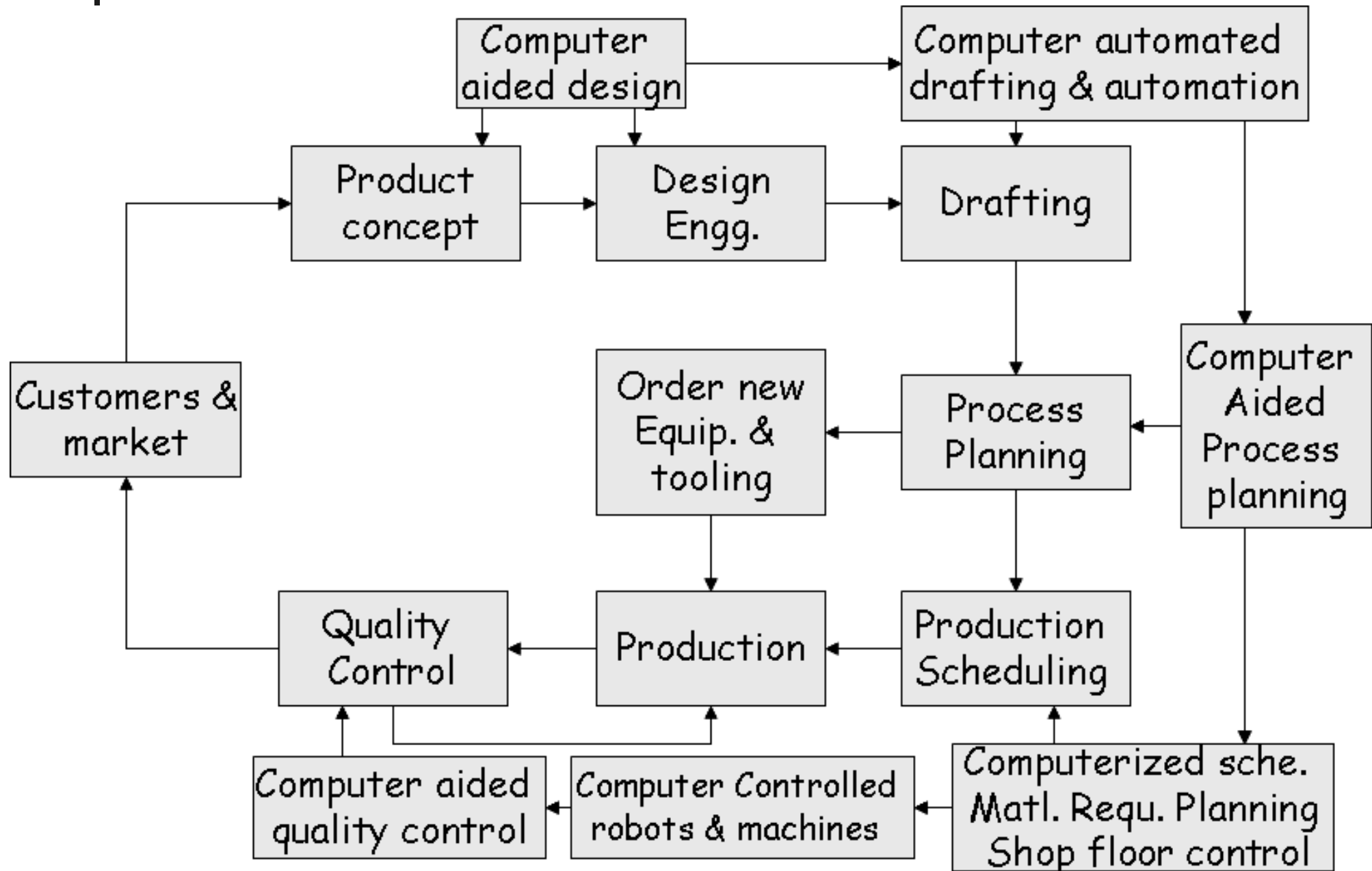
- Computer aided design and computer automated drafting are utilized in **conceptualization**, **design** and **documentation** of the product.
- Computers are used in process planning and scheduling to perform these functions ***more efficiently***.

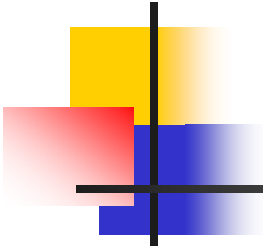


Product Cycle revised with CAD/CAM overlaid

- Computers are used in production to **monitor** and **control** the manufacturing operations.
- In quality control, computer are used to perform **inspections** and **performance tests** on the product and its components

Product Cycle revised with CAD/CAM overlaid





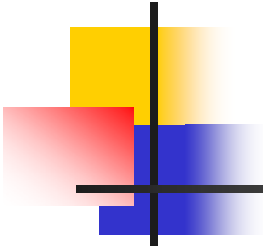
Mathematical Model for product life cycle

- Let **T1** be the time required to produce 1 unit of product.

This includes sum of individual process times plus time to assemble, inspect and package a single product.

- **T2** be time associated with planning and setting for each batch of production

This include the ordering of raw materials by purchasing department, time required in production planning to schedule the batch, setup times for each operation etc.

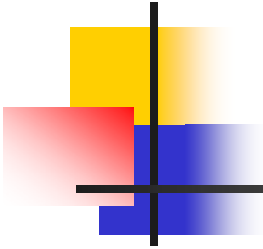


Mathematical Model for product life cycle, cont.

- **T3** be the time required for designing the product and for all other activities that are accomplished once for each different product.

These include process planning, cost estimating and pricing, building of special tools and fixtures.

- **B** be the number of batches produced throughout the product life cycle.
- **Q** be the number of units produced in each batch.



Mathematical Model for product life cycle, cont.

- The aggregate time spent on the product throughout its life cycle can be:

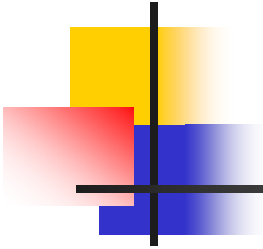
$$TT_{LC} = BQT_1 + BT_2 + T_3$$

Where TT_{LC} is the total time during the product life cycle

- The average time spent on each unit of product during its life cycle:

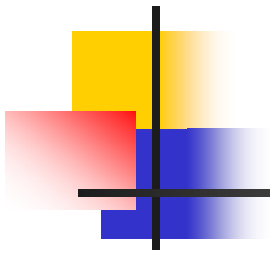
$$T_{LC} = T_1 + \frac{T_2}{Q} + \frac{T_3}{BQ}$$

Where T_{LC} = Average time

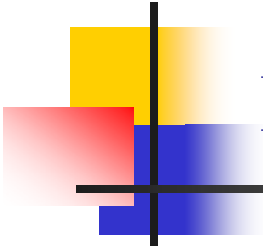


Mathematical Model for product life cycle, cont.

- In mass and batch production the T_2 and T_3 terms can be spread over a large number of units.
- Their relative values, therefore becomes less important as the production quantities increase. The T_1 term becomes most important term.
- In job shop manufacturing, the T_2 and T_3 terms can become significant because the quantities are so low.

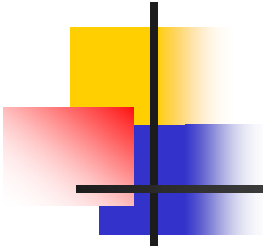


RELATIONSHIP BETWEEN CAD/CAM & AUTOMATION



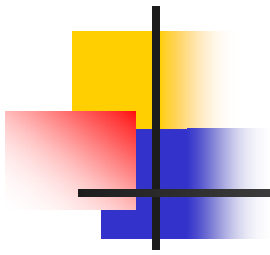
Relationship between CAD/CAM & Automation

- The goal of CAD/CAM and automation is to **reduce** the various *time elements* in product *life cycle*.
- With this there is increase in productivity and improve in standard of living.
- The automation technology is concerned with reducing the T1 and T2
- Elements, with emphasize on the unit production cost (T1).

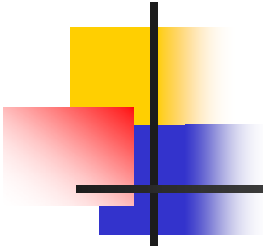


Relationship between CAD/CAM & Automation

- The CAD/CAM technology is concerned with all the **three terms** but is perhaps focused on the **T2** and **T3** terms in the life cycle model.
- CAD/CAM has made important contribution towards integrating the functions of design and manufacturing



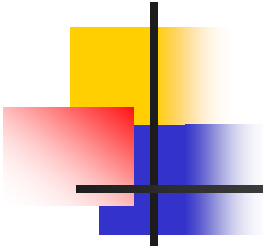
THE MANUFACTURING DATABASE FROM CAD/CAM



The manufacturing database from CAD/CAM

In the conventional manufacturing cycle practiced for so many years in industry:

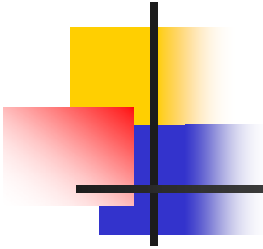
- Engineering drawings were Prepared by design draftsman and then used by manufacturing engineer to prepare process plan (“Route Sheet”)
- The activities involved in the designing the product were separated from the activities associated with process planning.
- Essentially a two-step procedure was employed.
- *This was both time consuming and involved duplication of efforts by design and manufacturing personnel*



The manufacturing database from CAD/CAM

In an integrated CAD/CAM system:

- A direct link is established between product design and manufacturing.
- It is goal of CAD/CAM not only to automate certain design phases of design and certain phases of mfg., but also to automate the transition from design to manufacturing.

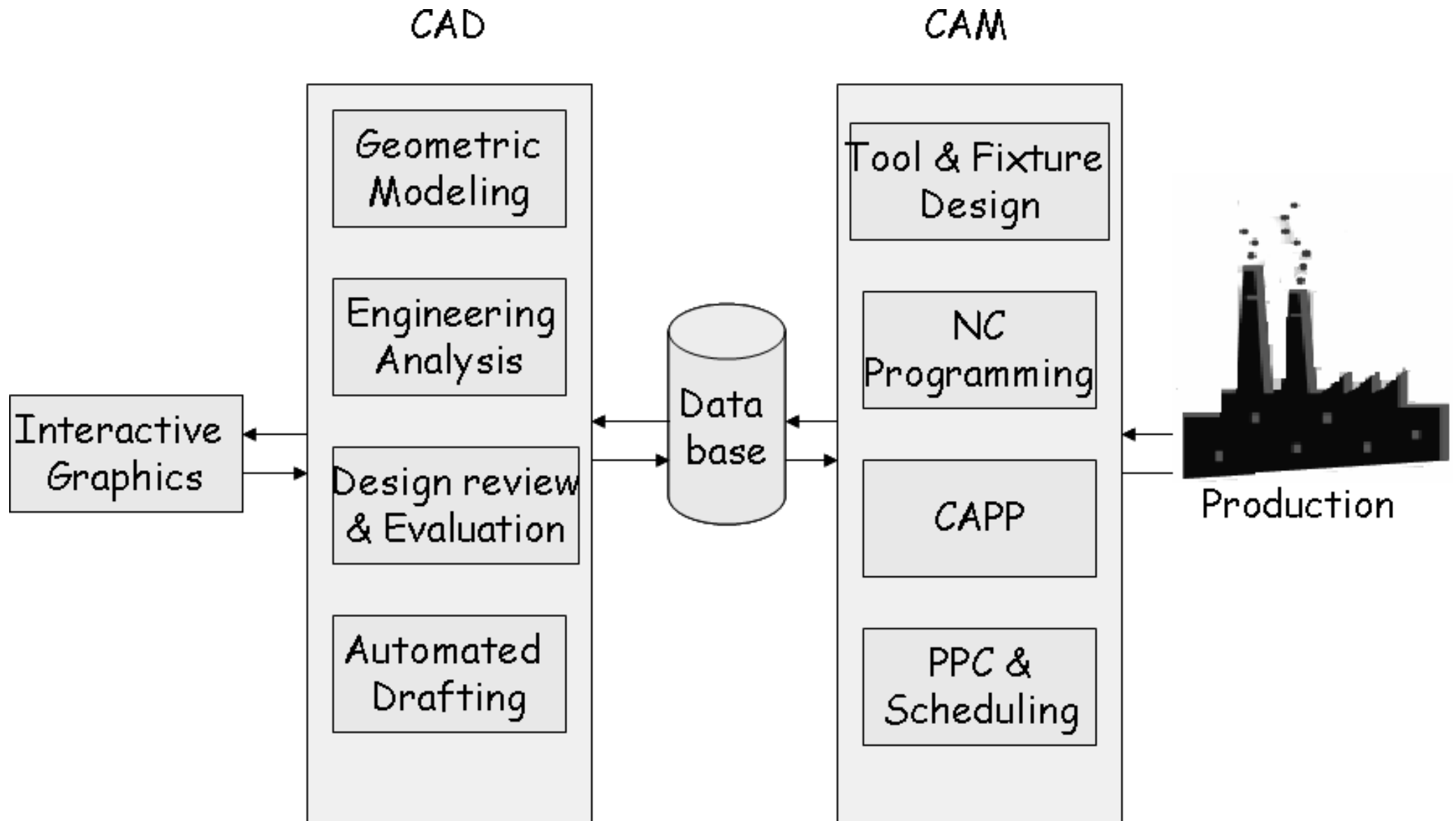


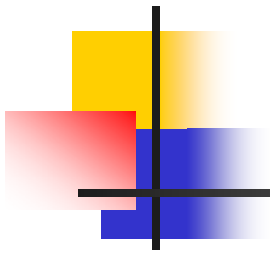
The manufacturing database from CAD/CAM

In an integrated CAD/CAM system:

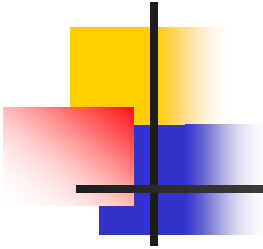
- Computer- based systems have been developed which create much of the data and documentation required to plan and manage the mfg. operations for the product.
- The manufacturing data base is an integrated CAD/CAM data base.
- It includes all the data on the product generated during design (geometry data, BOM , material specifications) as well as additional data required for mfg., much of which based on the product design.

The manufacturing database from CAD/CAM



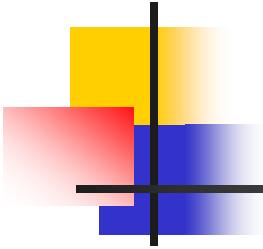


BENEFITS OF CAD



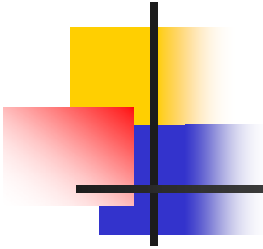
Benefits of CAD

- Productivity improvement in Design
- Shorter lead times
- Reduced engineering personnel requirement
- Customer modifications are easier to make
- Rapid response of the design analysis
- Improved accuracy of design



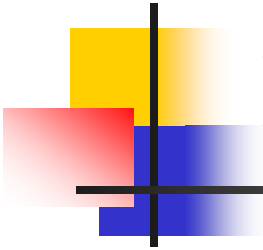
Benefits of CAD, cont.

- Avoidance of subcontracting to meet schedules
- Provide better functional analysis to reduce prototype testing
- Assistance in preparation of documentation
- Minimized transcription errors



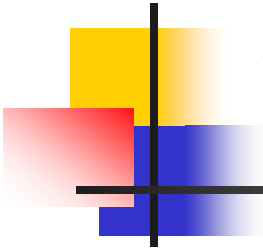
Benefits of CAD, cont.

- Designs have more standardization
- Fewer errors in NC part programming
- Better communication interfaces and greater understanding among engineers, designers, drafters, management and different project groups.



Definition 3

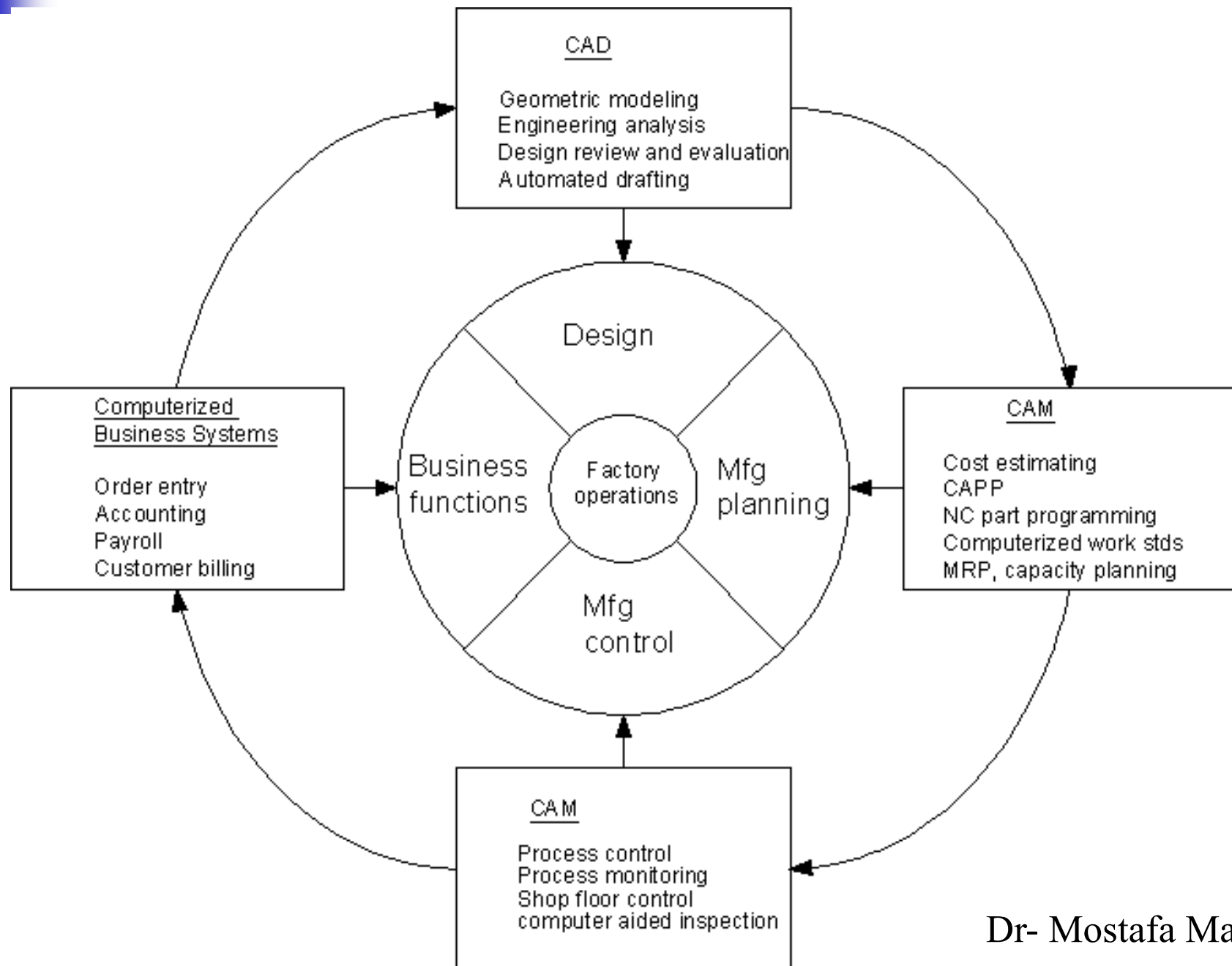
- **Computer Integrated manufacturing (CIM)** is the integration of total manufacturing enterprise by using **integrated systems** and **data communication** coupled with new managerial philosophies that improve organizational and personnel *efficiency*.



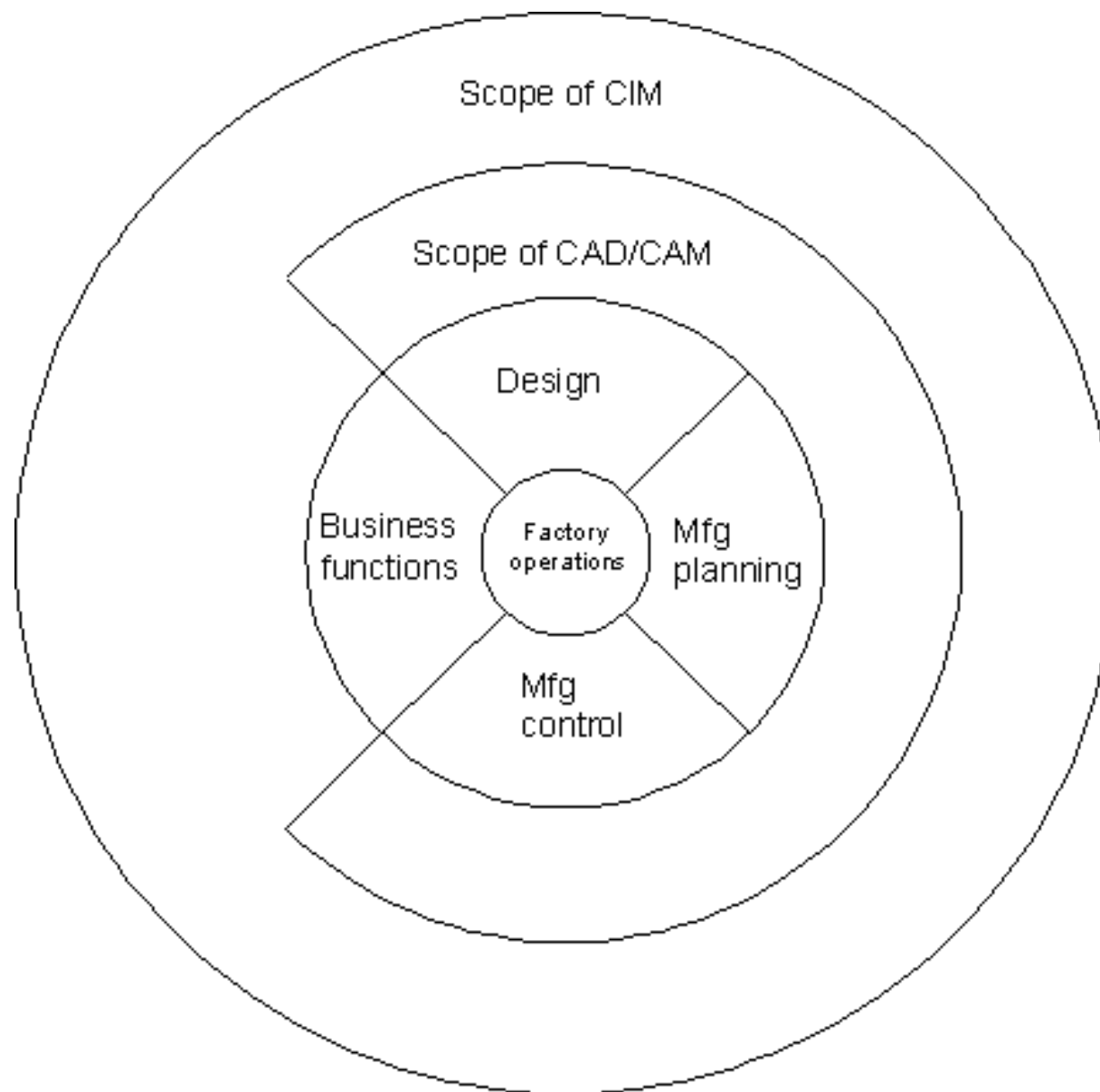
Definition 3, cont.

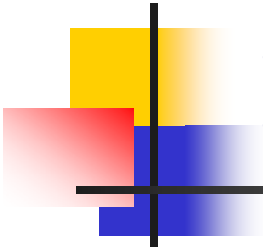
- **Computer-integrated manufacturing (CIM)** is manufacturing supported by computers. It is the total integration of Computer Aided Design / Manufacturing and also other business operations and databases.

Computerized elements of CIM system



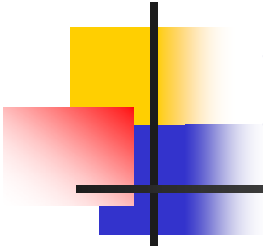
Scope of CAD/CAM and CIM





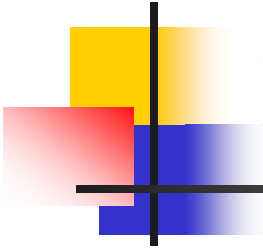
Definition 4

- **Computer-Aided Engineering (CAE)** Computer-aided Engineering analysis (often referred to as **CAE**) is the application of computer software in engineering to analyze the robustness and performance of components and assemblies.



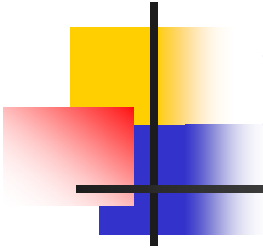
Definition 4, cont.

- It includes *simulation*, *validation* and *optimization* of products and manufacturing tools.
- Parts and assemblies designed in CAD software can be “analyzed” for their field performances right on the computer screen.
- Most of the olden-day **destructive testing** methods have found mathematical replacements in the modern-day CAE software.



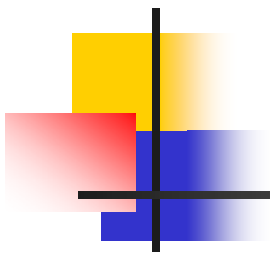
Definition 4, cont.

- CAE software delivers results that help analyzing designs.
- **Analysis tools are available for**
 - static stress-strain,
 - deflection,
 - thermal,
 - flow,
 - motion,
 - vibration

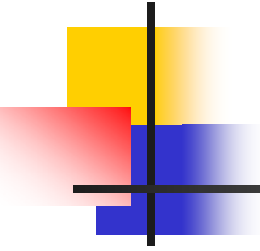


Definition 4, cont.

- This allows designers to “**design-right-the-first-time**”.
- Use of computer systems to analyze CAD geometry
- Allows designer to simulate and study how the product will behave, allowing for optimization
- Finite-element method (**FEM**)
- Divides model into interconnected elements
- Solves continuous field problems

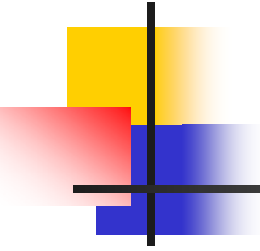


EXERCISE_PART B (5-10 MINS)



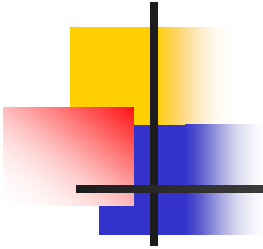
Exercise 1

- **Is it always good to use a CAD/CAM package?
Why?**



Exercise 2

- **What are the advantages of using a CAD/CAM system?**



How do CAD/CAM systems work?

Developing NC code requires an understanding of:

1. Part geometry
2. Tooling
3. Process plans
4. Tolerances
5. Fixtures

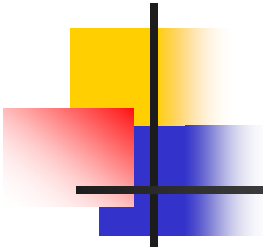
Most CAD/CAM systems provide access to:

1. Part geometry
2. Tooling

Instructions can be generated for a generic NC machine

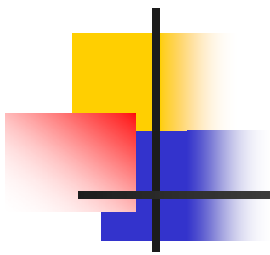
- A set of tool paths and positions can be automatically generated
- These paths can be edited and modified
- These paths and instructions can then be “posted” to a specific machine



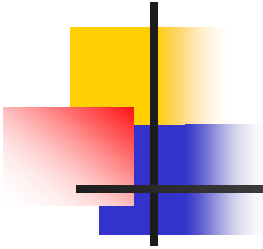


CAD/CAM Support

- AutoCAD
- Pro Engineer
- MasterCAM

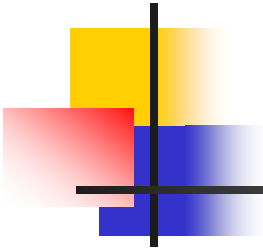


MASTERCAM



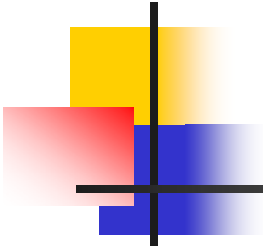
What do I need to begin MasterCAM?

- Part geometry
 - Draw or import
- Tooling
 - Library or create
- Process plans
- Fixtures
 - Define orientation and location



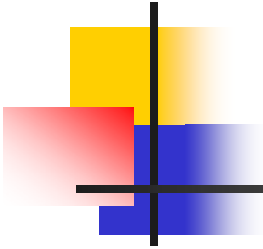
MasterCAM

- Mastercam is a three-dimensional geometry creation engine along with features to aid in tool path generation and verification.
- MasterCAM allows tool path planning and NC code generation for a given part. This part can either be drawn in MasterCAM or imported from other CAD packages



MasterCAM Drawing

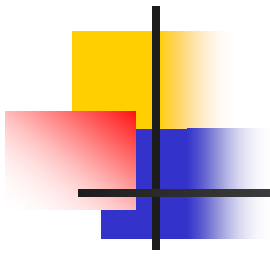
- Geometrical part drawing
 - In-built CAD package
- Two-dimensional parts
- Three-dimensional parts
 - Translators (include)
- IGES (international Graphics Exchange Standard)
- DXF (AutoCad)
- CADL (CADKey)



Tool Path Generation using MasterCAM

- **Tool path generation:**

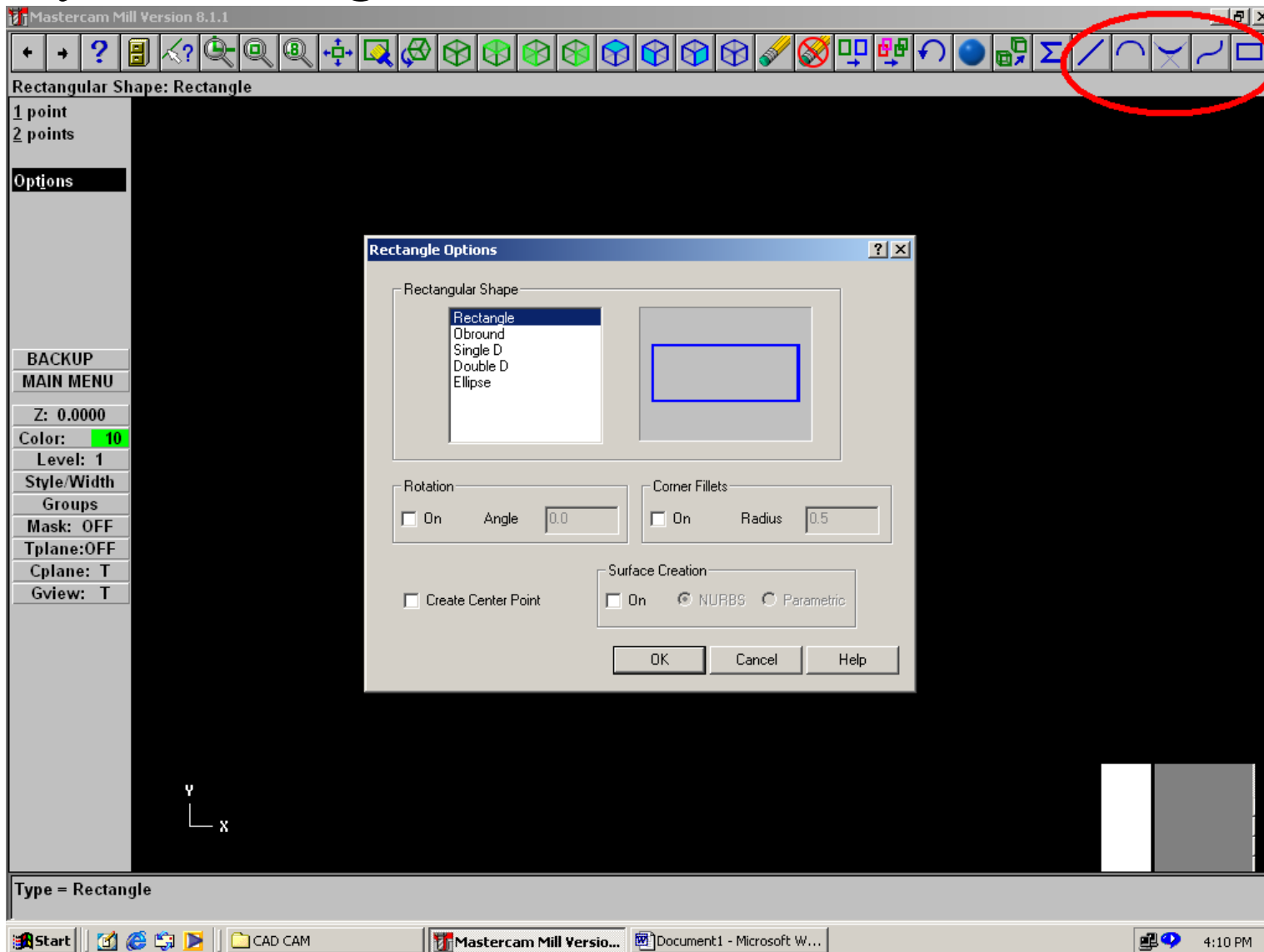
- Extensive Tool library
- Machining parameter selection
- NC program generator
- Animation to visualize machining operations



GETTING STARTED WITH 2D DRAWING

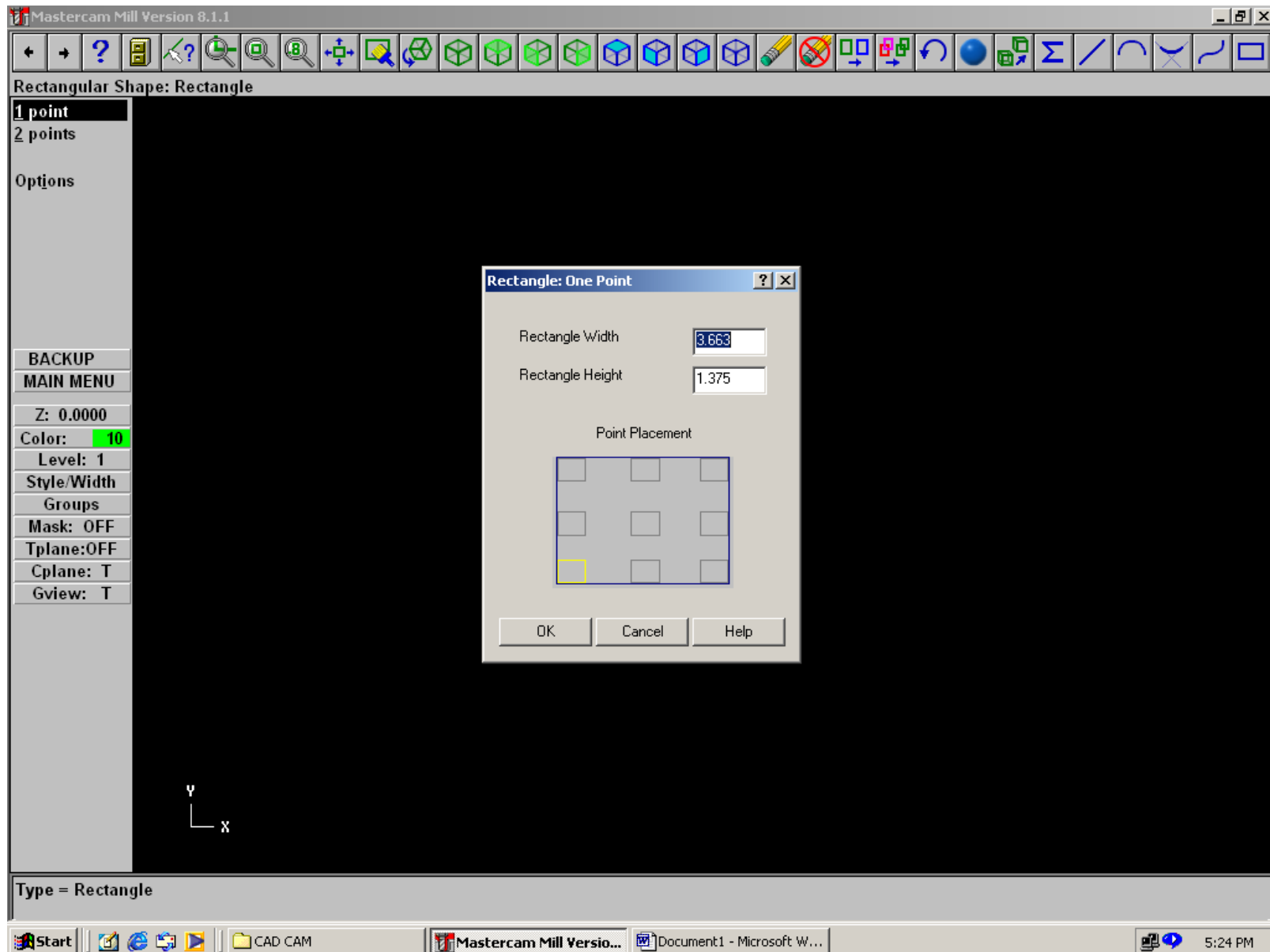
Getting Started with 2D Drawing

- Create simple 2D Geometries using basic shapes say a Rectangle



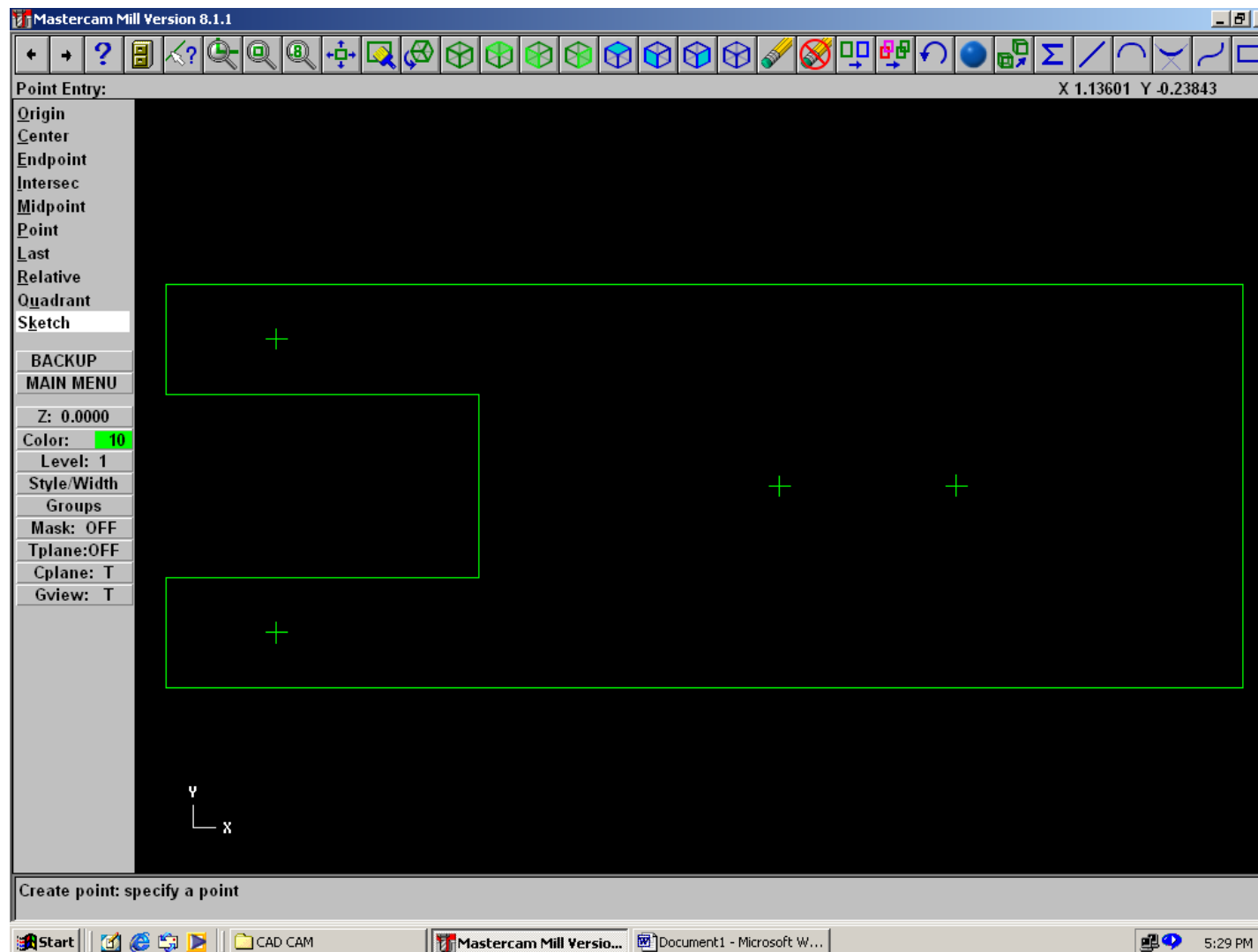
Building 2D Geometry

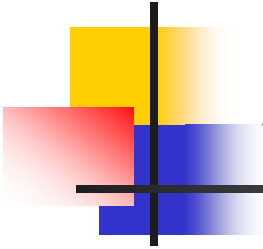
- Place and Dimension the Shape



Creating Shape with Lines and Rectangles

- Complete the Basic profile. We will now add arcs, fillets and trim entities





Adding Arcs, Radius and Fillets

- Click on Main Menu [Create- Arc-pt dia cir] on to dimension and place arcs/circles
- Click on Main Menu [Create-Fillets] to create Fillets. Dimension them Suitably.



Main Menu:

Screen-Repaint

Analyze
Create
File
Modify
Xform
Delete
Screen
Solids
Toolpaths
NC utils

BACKUP

MAIN MENU

Z: 0.0000

Color: 10

Level: 1

Style/Width

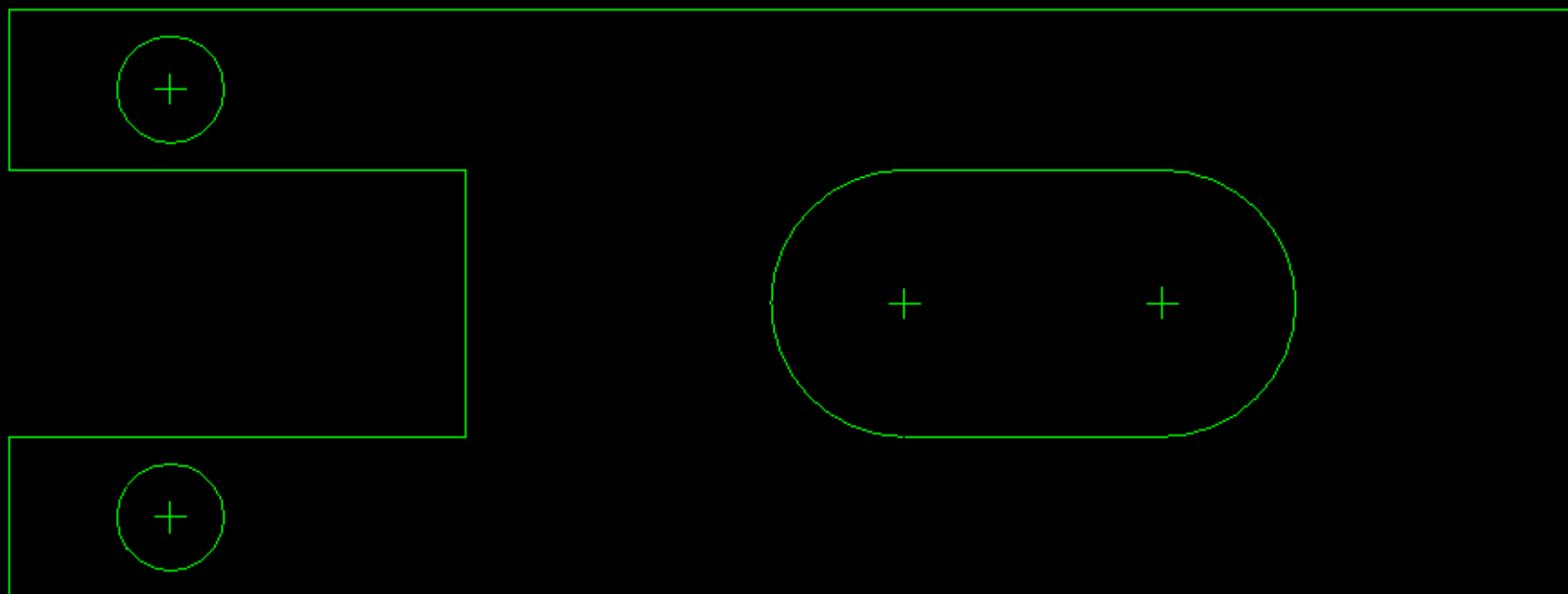
Groups

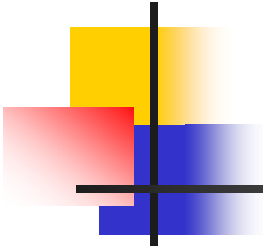
Mask: OFF

Tplane:OFF

Cplane: T

Gview: T





Completing our Mock Profile

- To Trim or Cut entities Main Menu [Modify-Trim-2 Entities] to get required profile.

Don't forget to Save your file



Delete: Select an entity or:

X 2.04936 Y -0.594

Chain
WindowArea
Only
All
Group
Result
Duplicate
Undelete

BACKUP

MAIN MENU

Z: 0.0000

Color: 10

Level: 1

Style/Width

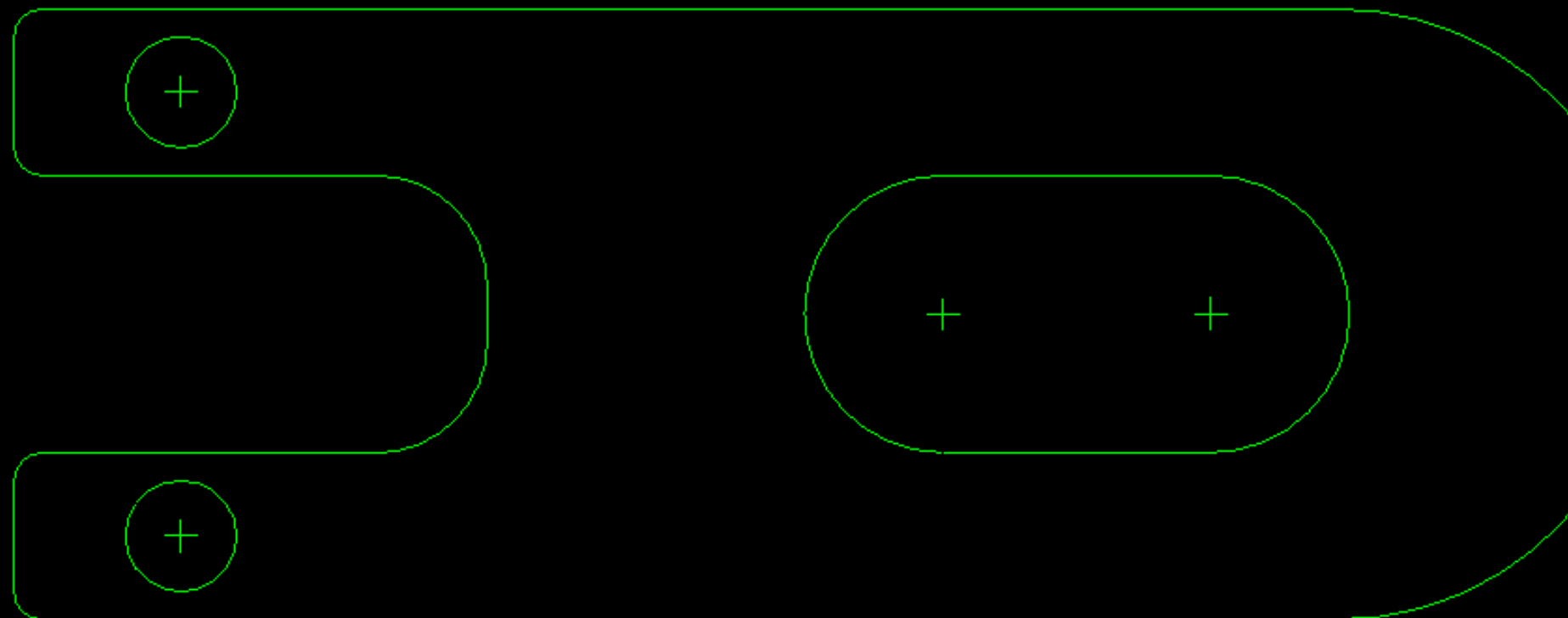
Groups

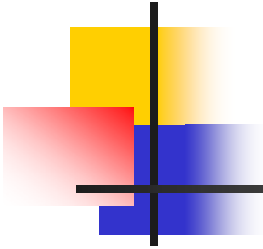
Mask: OFF

Tplane:OFF

Cplane: T

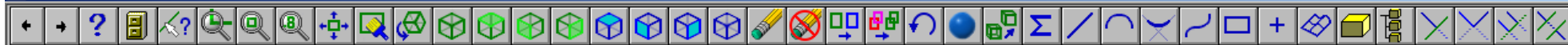
Gview: T





Getting started with Toolpaths

- Click on Main Menu [Toolpaths].
- MasterCAM lists the different machining operations



Toolpaths:

- New
- Contour
- Drill
- Pocket
- Face
- Surface
- Multiaxis
- Operations
- Job setup
- Next menu

BACKUP

MAIN MENU

Z: 0.0000

Color: 138

Level: 1

Style/Width

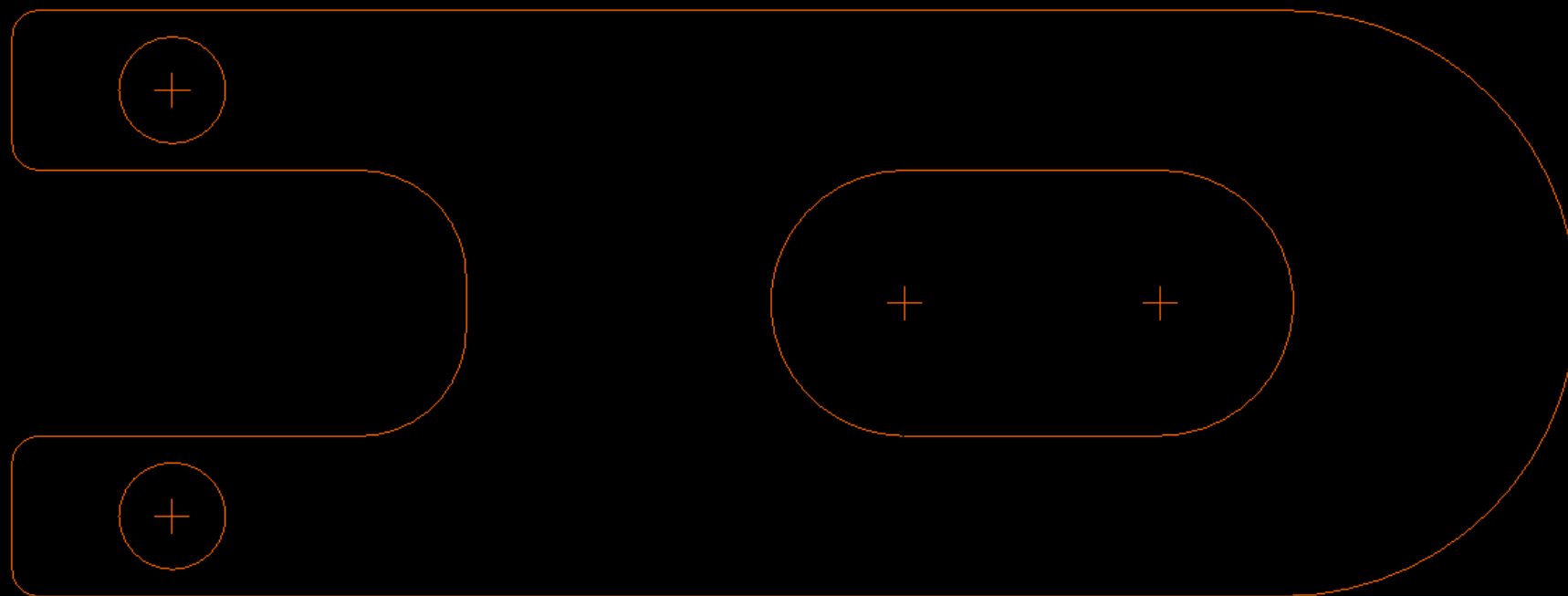
Groups

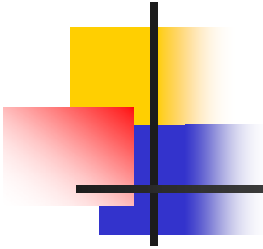
Mask: OFF

Tplane:OFF

Cplane: T

Gview: T





Contouring Options

- By Selecting Contour the various Contouring Options are listed.
- Select Chain and the geometry chains up and shows tool travel direction

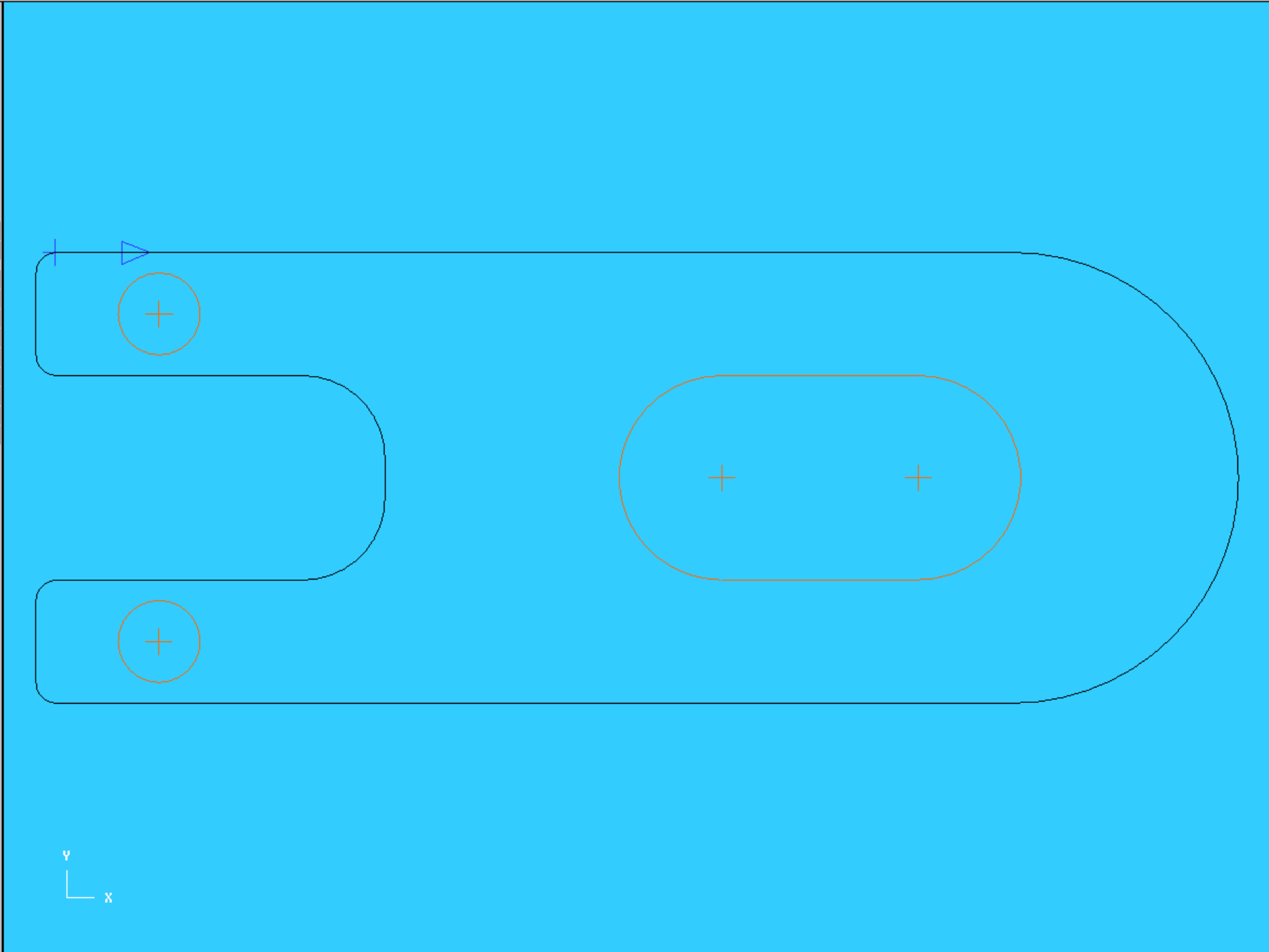


Contour: select chain 2

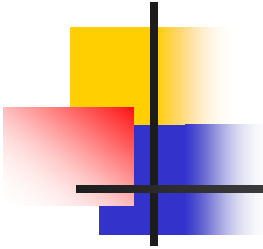
X 1.38595 Y 0.30420

Mode
Options
Partial
Reverse
Move fwd
Move back
Select strt
Unselect
Done

BACKUP
MAIN MENU
Z: 0.0000
Color: 138
Level: 1
Style/Width
Groups
Mask: OFF
Tplane: OFF
Cplane: T
Gview: T



Chaining mode: Full Chaining mask: None



Defining Tool Parameters

- Select the Contour type and Tool Parameters Window pops up.
- Feed the Right Parameters and Right Click to Select Specific Tools.

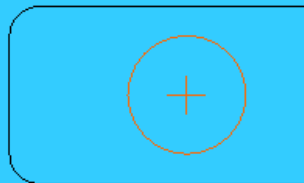
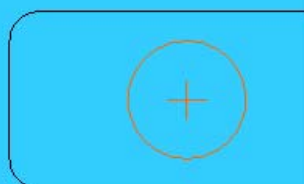


Contour: select chain 2

X -0.07847 Y 0.62853

Mode
Options
Partial
Reverse
Move fwd
Move back
Select strt
Unselect
Done

BACKUP
MAIN MENU
Z: 0.0000
Color: 138
Level: 1
Style/Width
Groups
Mask: OFF
Tplane: T
Cplane: T
Gview: T



Contour (2D) - C:\MCAM8\MILL\NCI\LESSON1.NCI - MPFAN

Tool parameters | Contour parameters

Left 'click' on tool to select; right 'click' to edit or define new tool

Tool #	1	Tool name		Tool dia	0.5	Corner radius	0.0
Head #	-1	Feed rate	6.4176	Program #	0	Spindle speed	1069
Dia. offset	1	Plunge rate	6.4176	Seq. start	1	Coolant	Flood
Len. offset	1	Retract rate	6.4176	Seq. inc.	1		

Comment

☐ Home pos... ☐ Ref point... ☐ Misc. values...

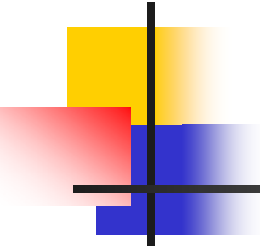
☐ Rotary axis... ☒ T/C plane... ☒ Tool display...

☐ To batch ☐ Canned text...

Change NCI...

OK Cancel Help

Chaining mode: Full Chaining mask: None



Selecting the Tools

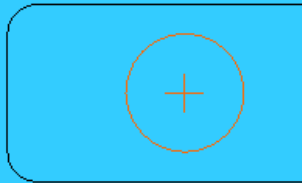
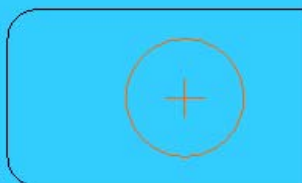
- MasterCAM has a whole range of tools in a tool library from which tool selections can be made. Now feed the correct parameters.
- You can also specify your own tools.



Contour: select chain 2

Mode
Options
Partial
Reverse
Move fwd
Move back
Select strt
Unselect
Done

BACKUP
MAIN MENU
Z: 0.0000
Color: 138
Level: 1
Style/Width
Groups
Mask: OFF
Tplane: T
Cplane: T
Gview: T



Contour (2D) - C:\MCAM8\MILL\NCI\LESSON1.NCI - MPFAN

Tool parameters | Contour parameters

Left 'click' on tool to select; right 'click' to edit or define new tool

#1- 0.5000
endmill1 flat

Tool #	1	Tool name	1/2 FLAT E	Tool dia	0.5	Corner radius	0.0
Head #	-1	Feed rate	6.4176	Program #	0	Spindle speed	1069
Dia. offset	41	Plunge rate	6.4176	Seq. start	100	Coolant	Off
Len. offset	1	Retract rate	6.4176	Seq. inc.	2		

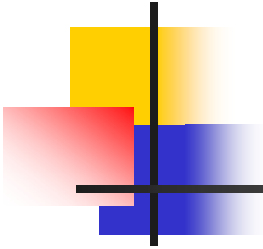
Comment
CONTOUR

☒ Home pos... ☐ Ref point... ☒ Misc. values...
☐ Rotary axis... ☒ T/C plane... ☒ Tool display...
☐ To batch ☐ Canned text...

Change NCI...

OK Cancel Help

Chaining mode: Full Chaining mask: None



Displaying Toolpaths

- Once both the Tool parameters and Contour Parameters have been Correctly defined.
- Click Ok to display the Toolpath.



Toolpaths:

X 3.44580 Y -0.40910

New
Contour
Drill
Pocket
Face
Surface
Multiaxis
Operations
Job setup
Next menu

BACKUP

MAIN MENU

Z: 0.0000

Color: 183

Level: 1

Style/Width

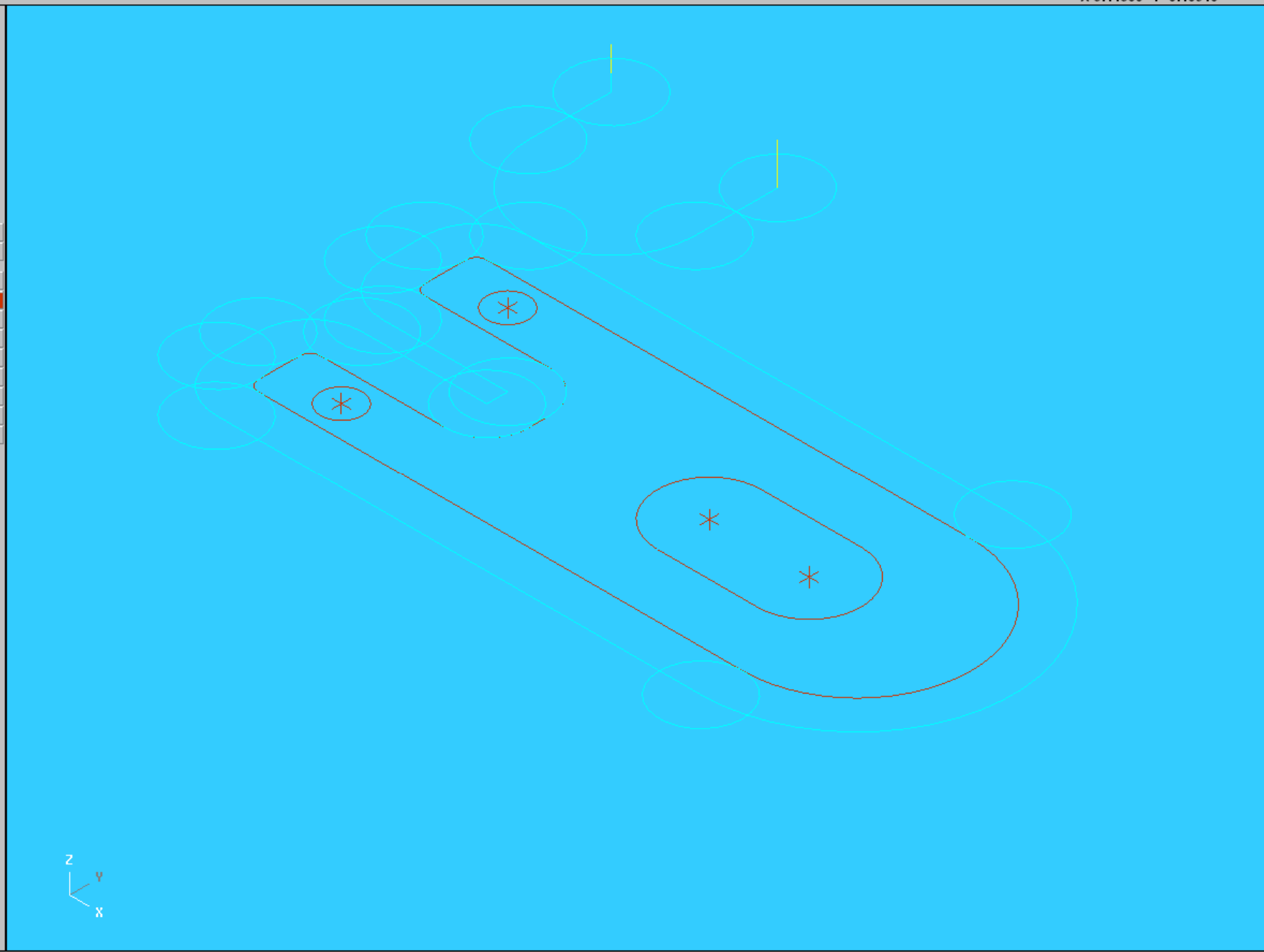
Groups

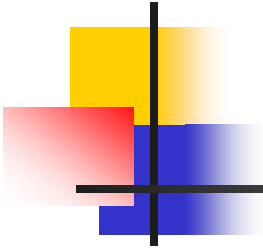
Mask: OFF

Tplane: OFF

Cplane: T

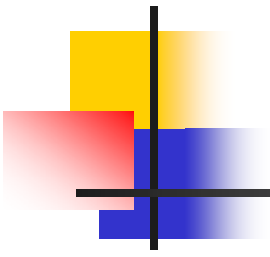
Gview: I



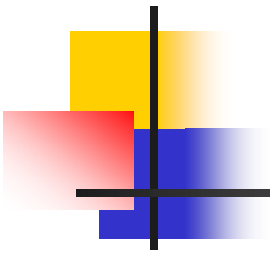


Toolpath Optimization?

- *MASTERCAM **does NOT** give a minimum time Toolpath.*
- *It gives the toolpath that has been selected.*



QUESTIONS



END